

Ayush Singh

AI Backend Engineer

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Education

Parul University, B.Tech in Computer Science

2022 – 2026

Experience

OpneEyes – AI Backend

May 26 - Ongoing

- Built enterprise-scale multimodal AI systems using RAG pipelines, improving ground-truth accuracy by **33%**
- Deploying ml model that unqiuly qgenerate question for enterprise level
- Deployed scalable AI infrastructure using Docker and AWS services for production environments.

Logicmotive – AI & Cloud Intern

Jan 26 – Apr 26

- Developed pipeline using AWS S3, Bedrock Knowledge Bases, Titan Embeddings, OpenSearch for search.
- Built scalable backend in Python + LangChain for retrieval pipelines and prompt orchestration
- Deployed using Docker (multi-stage builds + DockerSlim), AWS ECR, App Runner, reducing image size by **~60–70%** and improving deployment efficiency.

Skills

GenAI/AI: RAG, LangChain, LangGraph, LLMs, Embeddings, Vector Search, Prompt Engineering

Backend: FastAPI, REST APIs, Microservices

Cloud : AWS(EC2, S3, IAM, VPC, Lambda)

Devops: Docker, Kubernetes, Jenkins, Terraform, CI/CD, Linux, Git, GitHub Actions

Databases: MySQL, PineCone, PostgreSQL

Projects

Local Vendor Automation - AI/ML Developer (*Hackathon Top 10/1000, ₹5000 Prize*) [Github](#)

Tech Stack: Python, LangChain, Gemini API, Ollama, EasyOCR, Transfer Learning, RAG

- Built an AI onboarding automation system with **OCR PAN verification** and fraud checks..
- Developed **transfer learning image classifier** for auto product categorization..
- Built a FastAPI backend to connect OCR + classifier + RAG chatbot into one integrated workflow.
- Implemented RAG chatbot + safety filtering layer for grounded FAQ responses.

Production AI Agent [Live](#) | [Github](#)

Tech Stack: LangGraph, Langchain, Gemini, Grok, SQL, Docker, FastAPI, AWS, kubernetes

- Built and deployed an AI multi-agent platform using FastAPI, React, LangGraph, and Pinecone with scalable AI workflows.
- Optimized deployment using multi-stage Docker builds and slim images, reducing container size by ~60%.
- Automated AWS EC2 infrastructure with Terraform and integrated Amazon S3 for PDF storage and management.

E-commerce Hackathon Project [Live](#) | [Github](#)

Tech Stack: Docker, Kubernetes Jenkins, AWS-EC2, Linux

- Deployed a production-ready Online Shopping Portal using Docker, Kubernetes, Jenkins, and AWS EC2.
- Improved deployment speed by 50% and reduced Docker image size by 1GB using multi-stage builds.
- Automated setup and delivery using Bash, Docker Compose, and CI/CD.
- Hosted on AWS EC2 with scalable and reliable architecture.

ACHIEVEMENTS

- Top 10 / 1000 teams — PU Code Hackathon 3.0 (₹5000 Prize)
- Ranked Top 20% on LeetCode and Rank 5 among 9000 Students
- GFG 3-Star | CodeChef 2-Star